

YeEnat Weg



"The Way of the Mother" — Rooted in Ethiopian Tradition. Powered by AI.

Software Requirements Specification (SRS)

Field	Details
Document Title	SRS — YeEnat Weg Wellness Platform
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Status	Draft — For Review

1. Executive Summary

YeEnat Weg ("The Way of the Mother") is an AI-powered, Ethiopian-first nutrition and lifestyle management platform that integrates personalized meal planning, chronic disease management, and holistic lifestyle coaching — all grounded in indigenous Ethiopian food culture and health traditions.

Ethiopia faces a rising dual burden of malnutrition and non-communicable diseases (NCDs) such as Type 2 diabetes and hypertension, affecting millions of urban and peri-urban residents. Existing

wellness applications are overwhelmingly built for Western food systems, offering minimal relevance to Ethiopian users. YeEnat Weg bridges this critical gap.

The platform enables users to register their health profile (age, weight, height, medical conditions), receive an AI-generated personalized meal plan sourced entirely from local Ethiopian foods, log meals and track nutritional intake in real time, and receive lifestyle guidance — including sleep, hydration, exercise, and stress management routines — all tailored to an Ethiopian urban lifestyle.

The product is positioned as a freemium B2C SaaS application with supplementary B2B revenue from corporate wellness programs and anonymized health insights sold to hospitals, insurers, and research institutions — creating a compelling, scalable revenue model suitable for Series A investment.

2. Problem Statement & Market Opportunity

2.1 The Problem

Millions of Ethiopians managing conditions like diabetes, hypertension, and obesity receive generic dietary advice that does not account for local foods such as injera, teff, berbere, or enset. The result: poor adherence, worsening outcomes, and a healthcare system under strain.

- No nutrition app database covers Ethiopian staple foods comprehensively.
- Dietitians are scarce (Ethiopia has ~1 per 100,000 people); most advice is informal.
- Existing wellness apps are in English or foreign languages, creating adoption barriers.
- People with diabetes or hypertension have no affordable digital tool to manage diet and lifestyle holistically.

2.2 Market Opportunity

Market Segment	Estimated Size / Insight
Adults with Diabetes in Ethiopia	~4.3 million (IDF 2023); largely unmanaged
Urban Smartphone Users (Ethiopia)	~18 million and growing at 14% YoY
Corporate Wellness Market (Africa)	USD 2.1B projected by 2028
Health App Revenue (Sub-Saharan Africa)	USD 890M by 2027 (Statista)
Target Early Adopters	Urban professionals aged 25–50 managing NCDs

2.3 Unique Value Proposition

- Ethiopian food database with 500+ local foods and recipes, mapped to nutritional values.
- AI meal planner that respects fasting seasons (e.g., Ethiopian Orthodox fasting periods) and cultural dietary norms.
- Amharic-first interface, with support for Oromiffa and Tigrinya in future versions.
- Disease-aware personalization: specific nutritional guardrails for diabetes and hypertension.
- Lifestyle integration: sleep, exercise, and mental wellness modules built for Ethiopian urban schedules.

3. Product Overview

3.1 Product Name & Positioning

YeEnat Weg is positioned as "Your Personal Ethiopian Health Companion" — a mobile-first application (Android priority, iOS Phase 2) that feels culturally familiar while delivering clinically meaningful wellness support.

3.2 System Architecture Overview

3.2.1 High-Level Components

- Mobile Client: React Native application (Android & iOS)
- Backend API: Node.js/Python FastAPI — RESTful microservices
- AI/ML Engine: LLM-based meal planning + nutrient analysis pipeline
- Ethiopian Food Database: Custom-curated PostgreSQL database with USDA + local extension
- Notification & Reminder Service: Firebase Cloud Messaging
- Analytics & Insights Service: For B2B reporting layer
- Admin Dashboard: Web-based (React) for content management and business intelligence

3.2.2 System Context Diagram (Described)

The user interacts with the mobile app. The app communicates with the Backend API over HTTPS. The Backend calls the AI Engine for meal plan generation, queries the Ethiopian Food Database for nutritional data, and the Notification Service for reminders. A separate Analytics Service aggregates anonymized data for B2B reporting.

3.3 User Personas

Persona	Profile	Primary Need
Tigist, 38	Urban professional with Type 2 Diabetes. Uses smartphone daily.	AI meal plan avoiding high-glycemic foods; Amharic interface.

Persona	Profile	Primary Need
Dawit, 45	Government employee with hypertension. Limited nutrition knowledge.	Easy daily meal logging; blood pressure-friendly recipes from local foods.
Sara, 27	Health-conscious fitness enthusiast in Addis Ababa.	Personalized macro tracking; Ethiopian foods in workout nutrition plan.
Dr. Yonas	Corporate HR manager at a mid-size company.	Bulk employee wellness subscriptions; aggregate health reports.
Bethlehem, 55	Diabetic mother managing family nutrition.	Family meal planner; culturally appropriate fasting period guidance.

4. Functional Requirements

4.1 Module 1: User Registration & Health Profile

4.1.1 User Onboarding Flow

The onboarding flow collects the minimum data necessary to generate a personalized health plan. It is designed to be completable in under 4 minutes.

1. Account Creation: Phone number (OTP-verified) or Google Sign-In. Name, age, sex.
2. Physical Metrics: Current weight (kg), height (cm), target weight if applicable.
3. Medical History: Existing conditions (diabetes Type 1/2, hypertension, high cholesterol, kidney disease, pregnancy). Medications (optional, tagged for nutrient interaction alerts).
4. Dietary Preferences: Food allergies/intolerances (e.g., gluten, lactose). Fasting practices (Ethiopian Orthodox, Muslim Ramadan, custom). Vegetarian/vegan flag.
5. Lifestyle Inputs: Activity level (sedentary, light, moderate, active). Wake/sleep typical times. Work schedule type (office, field, shift).
6. Goals: Weight loss, weight gain, maintain weight, manage condition, improve energy.

4.1.2 Health Profile Requirements

Field	Requirement
BMI Auto-Calculation	Computed from height and weight; displayed with Ethiopian context messaging.
Caloric Needs	Harris-Benedict equation adjusted for activity level; shown to user.
Macro Split	AI-computed protein/carb/fat ratio based on conditions and goals.

Field	Requirement
Profile Editing	User can update any field at any time; plan regenerates automatically.
Data Privacy	All health data encrypted at rest (AES-256); user owns and can delete their data.

4.2 Module 2: AI-Powered Personalized Meal Planner

4.2.1 Meal Plan Generation

The core AI module generates a weekly meal plan for the user based on their health profile, goals, medical conditions, and cultural preferences. This is the flagship feature of YeEnat Weg.

- Generates a 7-day meal plan with 3 main meals and 1–2 snacks per day.
- All meals are sourced from the Ethiopian Food Database (local ingredients, local recipes).
- Respects fasting periods (no meat/dairy on fasting days for Orthodox users; halal compliance).
- Adjusts carbohydrate load for diabetic users (targets glycemic index < 55 for staple components).
- Limits sodium content for hypertension users (< 2,000 mg/day target).
- Accounts for seasonal food availability in Ethiopian markets.
- Plan regenerates weekly; user can request regeneration or swap individual meals.

4.2.2 Meal Plan Display & Interaction

- Daily view: breakfast, lunch, dinner, snack(s) displayed with images.
- Each meal shows: ingredients list, estimated preparation time, calories, key nutrients (protein, carbs, fat, fiber, sodium).
- "Replace This Meal" button: offers 3 alternative meals meeting the same nutritional constraints.
- "Shopping List" generation: aggregates weekly ingredients with estimated market quantities.
- Portion size guidance: shown in both grams and familiar Ethiopian household measures (e.g., "1 medium injera", "1 cup of shiro").

4.2.3 Ethiopian Food Database

Database Requirement	Specification
Minimum Food Entries (Launch)	500+ Ethiopian foods and dishes
Data Points per Food	Calories, protein, carbs, fat, fiber, sodium, glycemic index, vitamins (A, C, B12), iron, calcium
Recipe Database	200+ traditional recipes with ingredient-level breakdown
Fasting Compliance Tag	Boolean flag: suitable for Ethiopian Orthodox fasting

Database Requirement	Specification
Regional Tagging	Oromia, Amhara, Tigray, Sidama, Somali regional dishes
Data Source	Ethiopian Public Health Institute data + WHO + manual dietitian review

4.3 Module 3: Nutrition Tracker

4.3.1 Meal Logging

Users log each meal consumed. The tracker computes daily and weekly nutritional intake against personalized targets.

- Log a meal from: (a) Planned meal — tap to mark as eaten, (b) Search Ethiopian food database, (c) Barcode scan for packaged foods (Phase 2), (d) Photo-based food recognition via AI (Phase 2).
- Portion adjustment: user can adjust default serving size via slider or numeric input.
- Quick-add common items: "1 injera", "1 cup tej", "1 tibs portion" etc.
- Water intake tracker integrated into the daily log.

4.3.2 Nutrition Dashboard

- Daily ring chart: calories consumed vs. target (color-coded green/amber/red).
- Macro breakdown bars: protein, carbs, fat — actual vs. target.
- Key micronutrient flags: alerts if sodium, sugar, or saturated fat exceeds safe thresholds.
- Weekly trend charts: 7-day caloric intake, weight trend (if logged), adherence score.
- "Nutritional debt" view: shows nutrients that are consistently under-consumed (e.g., iron, fiber).

4.3.3 Disease-Specific Tracking

Condition	Additional Tracking Features
Diabetes	Estimated Glycemic Load per meal; daily carbohydrate budget with red-line alert; sugar intake gauge; meal timing reminders (consistent timing reduces spikes).
Hypertension	Daily sodium tracker with strict 2,000 mg ceiling; potassium intake monitoring (bananas, avocados, sweet potato promoted); alcohol flagging.
Weight Management	Caloric deficit/surplus tracking; BMI trend; weekly body weight log with chart.
General Wellness	Fiber intake, hydration, protein adequacy, micronutrient diversity score.

4.4 Module 4: Lifestyle Management

4.4.1 Sleep Routine

- User sets target sleep and wake times during onboarding.
- Smart notifications: "Wind down" reminder 30 minutes before target sleep time.
- Morning sleep log: user rates sleep quality (1–5 stars) and duration.
- AI insight: correlates poor sleep patterns with the following day's nutritional adherence.
- Ethiopian-specific: guidance respects early-rising norms and morning coffee (Buna) ceremony acknowledgment.

4.4.2 Exercise & Physical Activity

- Daily step goal (default: 7,500 steps; adjustable by user or auto-reduced for health conditions).
- Curated exercise library: categorized by home workouts, park/outdoor, gym, and low-impact (suitable for hypertensive or diabetic users).
- Weekly exercise plan: 3–5 sessions, duration 20–45 minutes, with warm-up and cool-down.
- Post-exercise calorie adjustment: nutrition tracker recalibrates remaining daily calories.
- Walking route suggestions: leverages GPS to suggest parks and open spaces in Ethiopian cities.

4.4.3 Hydration Tracking

- Daily water intake goal (default: 2.5L, adjusted for activity, pregnancy, climate).
- Hourly reminders with tap-to-log functionality ("I drank a cup").
- Tej, coffee (Buna), and local beverage tracking with hydration credit calculations.

4.4.4 Wellness Insights & AI Coaching

- Daily "health tip" personalized to the user's profile — sourced from Ethiopian health traditions and modern nutrition science.
- Weekly AI report: "Here is how you performed this week" — adherence score, wins, areas to improve.
- Streak system: consecutive days of meal logging, exercise, and sleep logging rewarded with badges.
- Behavioral nudges: contextual push notifications triggered by patterns (e.g., "You tend to skip breakfast on Mondays — here is a quick option.").

4.5 Module 5: Smart Notifications & Reminders

- Meal reminders: configurable breakfast, lunch, and dinner times.
- Medication reminders: user can add custom medication times (non-medical, reminder only).
- Meal plan refresh: weekly notification when new plan is ready.
- Motivational messages: Amharic-language motivational quotes relevant to health.
- Health alerts: if daily caloric intake is below 60% of target by 8 PM, gentle prompt.

4.6 Module 6: Community & Social Features (Differentiator)

Community features increase retention and create network effects that are difficult for competitors to replicate.

- "Wellness Circles": private groups (family, friends, colleagues) sharing meal logs and progress.
- Recipe sharing: users can post custom Ethiopian recipes for community rating and nutrition analysis.
- Weekly community challenge: e.g., "30-minute walk streak for 7 days" or "Sugar-free week."
- Leaderboard (opt-in): adherence score ranking within Wellness Circles.
- Community-reviewed "Eating Out" guide: healthy choices at popular Ethiopian restaurants and hotels.

5. Non-Functional Requirements

5.1 Performance

- Meal plan generation: < 10 seconds on first load; cached for instant subsequent access.
- App launch time: < 3 seconds on mid-range Android devices (Tecno, Samsung A-series).
- Food database search: results returned in < 500ms.
- App functions with intermittent connectivity: core tracking works offline, syncs when online.

5.2 Usability

- Primary language: Amharic. English toggle available.
- Minimum font size: 16px for all body content (accessibility for older users).
- Voice input: meal logging by speaking food names in Amharic (Phase 2).
- Designed for users with limited tech literacy: maximum 3 taps to complete any core action.

5.3 Security & Privacy

- Health data encrypted at rest (AES-256) and in transit (TLS 1.3).
- Ethiopian Personal Data Protection Proclamation (2022) compliance.
- GDPR-aligned data handling for international users.
- B2B data: fully anonymized and aggregated — no individually identifiable health data shared.
- User can export or permanently delete all data at any time.

5.4 Scalability

- Backend infrastructure: containerized (Docker/Kubernetes) to scale horizontally.
- Database: PostgreSQL with read replicas; Redis caching layer for food database queries.
- Target: 10,000 users at launch (Addis Ababa); 100,000 within 12 months.

6. Revenue Model & Business Case

6.1 Revenue Streams

Revenue Stream	Description & Projection
Freemium — Basic (Free)	Core meal logging, 7-day meal plan, basic lifestyle tips. Unlimited users. Drives acquisition.
Freemium — Premium (ETB 199/month)	Advanced AI coaching, disease-specific dashboards, community features, offline access, detailed analytics. Target: 8% conversion.
Corporate Wellness Packages	Per-employee subscription sold to companies (ETB 99/employee/month). HR dashboard with aggregate health reports. Target: 20 corporate clients in Year 1.
Healthcare Provider API	Licensed API for hospitals and clinics to prescribe YeEnat Weg as a dietary tool to patients. SaaS license fee.
Anonymized Health Data Insights	Aggregated, de-identified nutritional and lifestyle trend reports sold to FMCG companies, insurers, and research institutions (e.g., "Urban Ethiopian dietary patterns by age group").
Healthy Food Marketplace (Phase 3)	Commission on orders of curated healthy food baskets and local farmer produce delivered to users.

6.2 Financial Projections (Year 1)

Metric	Conservative	Optimistic
Registered Users (Month 12)	15,000	50,000
Premium Subscribers	1,200	4,000
Premium Revenue (ETB)	~2.9M / year	~9.6M / year
Corporate Clients	10	30
Corporate Revenue (ETB)	~1.2M / year	~3.6M / year
Total Year 1 Revenue (ETB)	~4.1M (~\$30K USD)	~13.2M (~\$96K USD)

Note: These projections assume a 6-month development period and 6-month go-to-market phase. USD equivalents at 137 ETB/USD.

7. Key User Stories

User Story	Acceptance Criteria
As Tigist (diabetic), I want a 7-day meal plan using Ethiopian foods I can cook at home.	Plan generated in < 10s. All meals sourced from local foods. Glycemic load flagged per meal. Fasting days respected.
As Dawit (hypertensive), I want to log my lunch and see if my sodium is within limits.	Food search returns results in < 1s. Daily sodium gauge updates immediately. Alert fires if > 1,800 mg.
As Sara, I want to swap a meal I don't like for something equivalent.	3 alternatives shown in < 2s, all meeting same caloric and macro constraints.
As any user, I want to receive a reminder to drink water every 2 hours.	Notification delivered within 5 minutes of scheduled time. Dismissible. Adjustable in settings.
As Dr. Yonas (HR manager), I want a monthly wellness report for my 200 employees.	Anonymized aggregate PDF report generated. No individual data exposed.
As Bethlehem, I want the app in Amharic so I can use it without difficulty.	100% of core user flows available in Amharic. No English-only barriers.

8. Development Roadmap

8.1 MVP Scope (Hackathon Demo — June 14, 2026)

The hackathon demo focuses on end-to-end flow demonstration of the core value proposition.

7. User registration and health profile collection (working form).
8. AI meal plan generation — 3-day sample plan using Ethiopian food database (minimum 100 foods).
9. Meal plan display with nutritional breakdown.
10. Meal logging: log 1 meal from database, see updated daily nutrition gauge.
11. Disease flag demo: show different plan for diabetic vs. non-diabetic profile.
12. Lifestyle module mockup: sleep and exercise recommendations displayed.
13. Revenue model slide and B2B dashboard wireframe.

8.2 Phase 1: Launch (Months 1–4 post-hackathon)

- Full 7-day meal planner with weekly refresh.
- Complete Ethiopian food database (500+ entries, dietitian-verified).
- Full nutrition tracker with daily and weekly dashboards.
- Sleep, exercise, and hydration lifestyle modules.
- Amharic UI complete. Android app published on Google Play.
- Freemium model live (Free + Premium tiers).

8.3 Phase 2: Growth (Months 5–9)

- iOS app launch.
- Barcode scanner for packaged foods.
- AI photo food recognition.
- Community/Wellness Circles feature.
- Corporate wellness portal and HR dashboard.
- Oromiffa language support.

8.4 Phase 3: Scale (Months 10–18)

- Healthcare provider API and clinical partnership program.
- Anonymized data insights B2B product launch.
- Healthy food marketplace integration.
- Expansion to Kenya and Tanzania (Swahili localization).

9. Risk Analysis & Mitigation

Risk	Impact	Mitigation
Ethiopian food nutritional data is incomplete or inaccurate.	High — core feature relies on this.	Partner with Ethiopian Public Health Institute; engage licensed dietitian for database review.
Low smartphone penetration in rural areas limits reach.	Medium — initial TAM reduced.	Target urban users first; design for offline use and feature phones in Phase 3.
User retention drops after first week (common in wellness apps).	High — threatens revenue model.	Gamification (streaks, badges), community features, and weekly AI reports drive habit formation.

Risk	Impact	Mitigation
Regulatory ambiguity around health data in Ethiopia.	Medium — legal risk.	Comply with PDPP (2022); avoid clinical diagnoses; position as wellness tool, not medical device.
Competition from global apps (MyFitnessPal, etc.) adding local foods.	Medium — commoditization risk.	Deep cultural specificity (Orthodox fasting, Amharic UX, local food seasonality) is hard to replicate fast.

10. Success Metrics & KPIs

KPI	Target (Month 12)
Registered Users	15,000+
Daily Active Users (DAU)	35% of registered users
7-Day Retention Rate	> 45%
Premium Conversion Rate	> 6%
Meal Plans Generated	50,000+ / month
Meals Logged per Active User	> 2 per day
App Store Rating	> 4.3 / 5
Corporate Clients	> 10
Net Promoter Score (NPS)	> 50

11. Appendix — Technology Stack Recommendation

Layer	Recommended Technology
Mobile Frontend	React Native (cross-platform Android/iOS from single codebase)
Backend API	Python FastAPI (performance, AI ecosystem compatibility)
Database	PostgreSQL (relational, JSONB for flexible food data)
AI/LLM Engine	OpenAI GPT-4o or Anthropic Claude API for meal plan generation

Layer	Recommended Technology
Cache Layer	Redis — food database query caching, session management
Authentication	Firebase Auth (OTP + Google Sign-In)
Push Notifications	Firebase Cloud Messaging (FCM)
File Storage	AWS S3 / Backblaze B2 (food images, user avatars)
Analytics	PostHog (product analytics, self-hostable)
Admin Dashboard	React + Recharts + shadcn/ui
CI/CD	GitHub Actions → Docker → AWS ECS or Railway.app
Monitoring	Sentry (errors), Grafana (metrics)

The technology stack prioritizes speed to market, developer familiarity with tools available in Ethiopia, and a cost-efficient hosting model suitable for a startup scaling from 0 to 100,000 users.

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